BTS-VII(SS)-04.16-0668 Reg. No. [ ]

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B.Tech. Degree VII Semester Special Supplementary Examination April 2016

ME 1703 MACHINE DESIGN II

(2012 Scheme)

(Use of Design Data handbook is permitted. Data not given may be suitably assumed)

Time : 3 Hours Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) How does the function of a brake differ from that of a clutch?

(b) Explain belt slip and creep in flat belt drives.

(c) Explain the bevel gear nomenclature, with a neat sketch.

(d) Derive the equation for beam strength of spur gear, with the help of a sketch.

(e) Explain wedge film lubrication and squeeze film lubrication.

(f) What are the desirable properties for effective bearing materials?

(g) Describe (i) tolerance (ii) surface finish.

(h) What are the points to be considered while designing a part which needs machining?

PART B

(4 × 15 = 60)

II. A single dry plate clutch is to be designed to transmit 10 kW at 1200 rpm. The ratio of the mean radius to the face width as 4, the intensity of pressure as 0.07 MPa and coefficient of friction as 0.25. Determine the diameter of shaft, face width, inner and outer radii of clutch plate. (15)

OR

III. Power of 75 kW at 900 rpm is to be transmitted from an electric motor to compressor shaft at 300 rpm by V-belts. The approximate larger pulley diameter and the centre distance are 1500 mm and 1650 mm, and the overload factor is to be taken as 1.25. A belt with cross-sectional area of 350 mm^2 and density 1000 kg/m^3 and having an allowable tensile strength 2 MPa is available for use. The coefficient of friction between the belt and the pulley may be taken as 0.28. The driven pulley is overhung to the extent of 300 mm from the nearest bearing and is mounted on a shaft. Design the belt drive and the shaft. (15)

IV. A gear drive is required to transmit a maximum power of 25 kW. The velocity ratio is 1:2 and the rpm of pinion is 250. The approximate centre distance between the shaft may be taken as 800 mm. Design the gear drive, also check and design for static and limiting wear load. (15)

OR

V. Design 20° involute worm and gear to transmit 15 kW with worm rotating at 1400 rpm and to obtain a speed reduction of 12:1. The distance between the shafts is 300 mm. (15)

VI. A 100 mm diameter shaft operating at 2000 rev/min is supported by means of a 0.15 m long full journal bearing which is supporting radial load of 66 kN. The oil temperature is to be limited to 60°C, the surrounding air temperature 38°C. Assume (Zn/P) = 28.5. Determine: (15)

(i) Co-efficient of friction.
(ii) Bearing pressure.
(iii) Heat generated.
(iv) Heat dissipated.
(v) Oil to be used.
(vi) Is artificial cooling required?

OR

VII. Select a suitable bearing, which is to operate at 1300 rpm and is acted upon by a 7 kN radial load and 4.5 kN thrust load. The inner ring rotates, the load is steady and the service is continuous. The shaft diameter is 40 mm and life expectancy is 600 hrs. If the bearing outer ring is rotating and is subjected to light shock, how the life of bearing changes? (15)

VIII. (a) What are the design conditions for forging? (7)
(b) What are the design recommendations for welded parts? (8)

OR

IX. What are the general design recommendations for castings? (15)

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